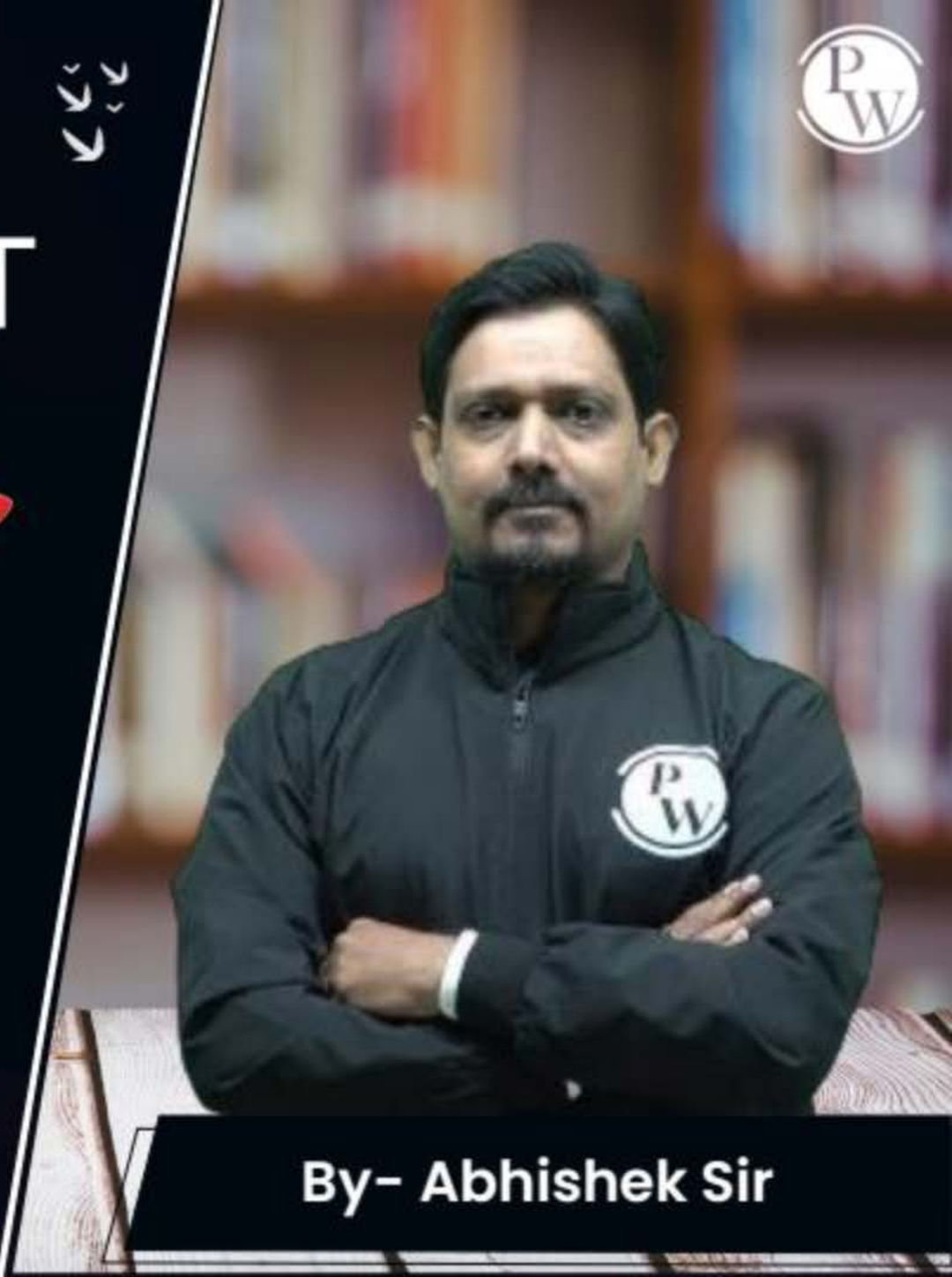
Computer Science & IT



Fundamental of C Language

Function & Storage Class

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



- Topic — global variable
- Topic — Local variable
- Topic — Nested function Call
- Topic
- Topic

Slide

Topics to be Covered



Topic — Memory Layout

Topic — Storage class : auto

Topic — Static

Topic — Extern

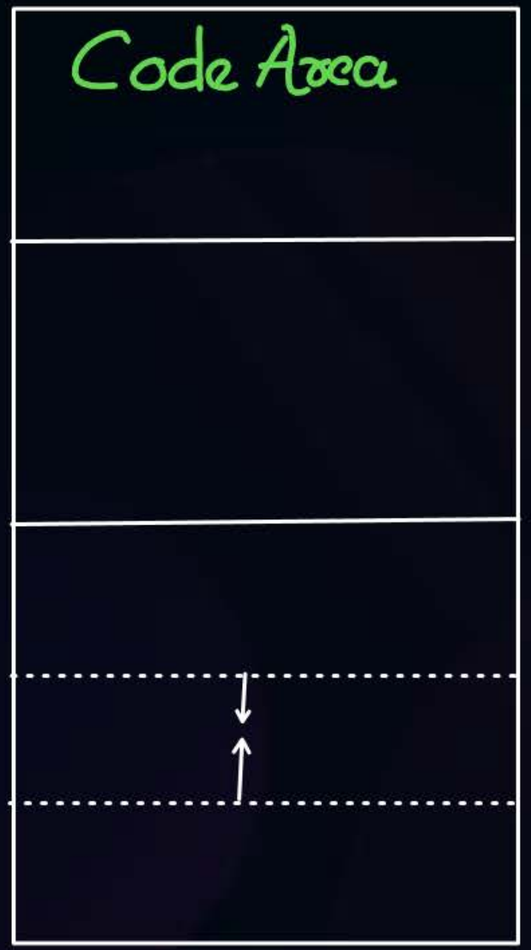
Topic — Register

Slide



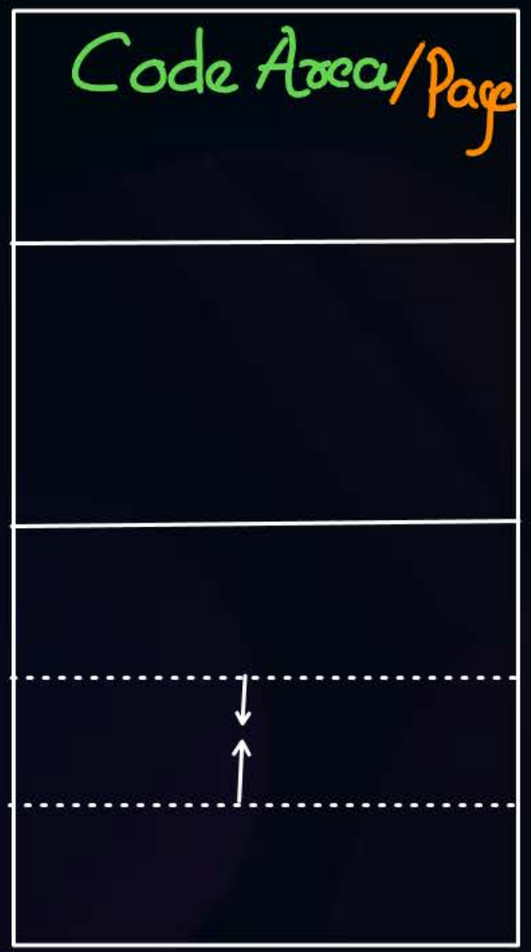
Storage class

Memory layout of C program :- When a program is in execution then it uses different sections of Memory explained as follows.



Slide

Storage class



Code Area :- Linker & Loader Loads executable file in this area for execution

Slide



Storage class



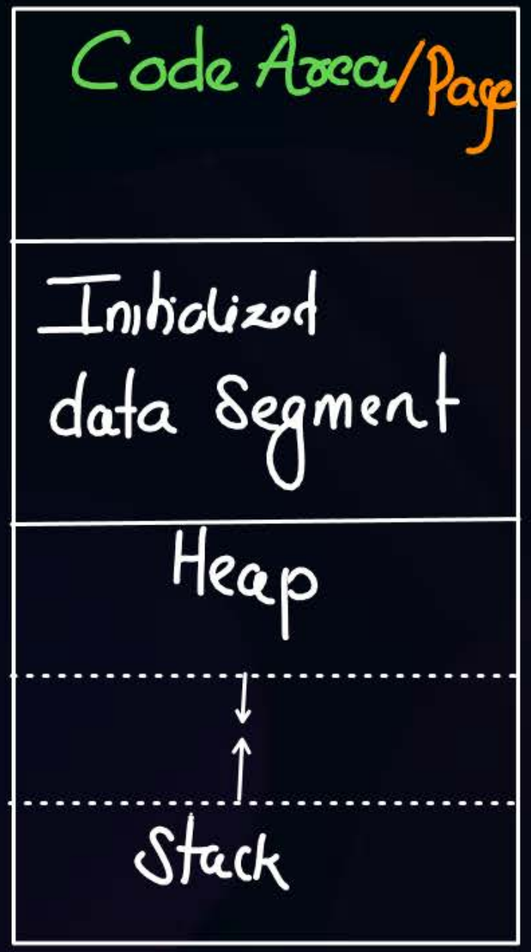
Initialized data Segment : All data which has has intial value defined stored in Initialized data Segment

global variable by default have initial value - 0.

Compile time allocation

Slide

Storage class



Heap :- global variable & Local variable

Memory allocated by compiler.

user can also allocate & deallocate the memory.

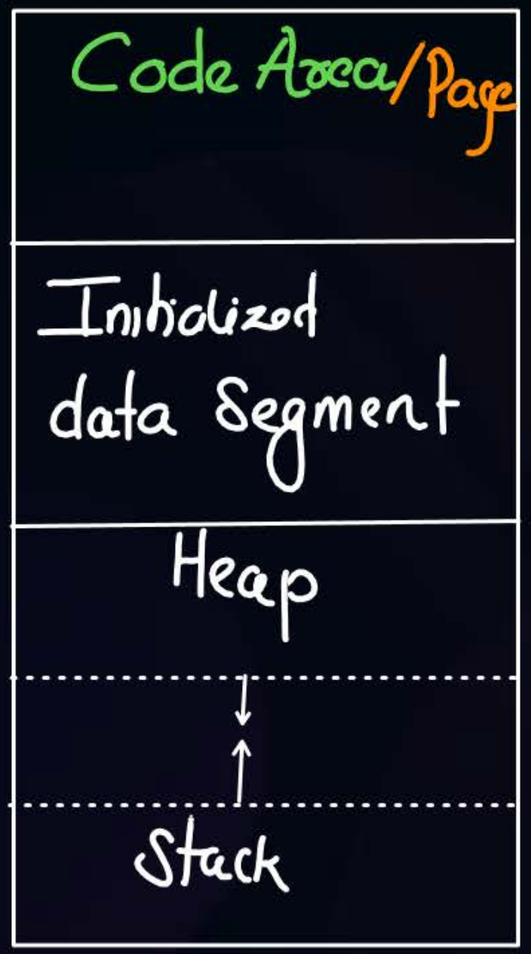
Run time Memory allocation (Dynamic)

Heap Area user can allocate Memory using malloc(). calloc(), free()

Slide



Storage class



Stack :- Memory allocated in stack is also Dynamically allocated i.e run-time.

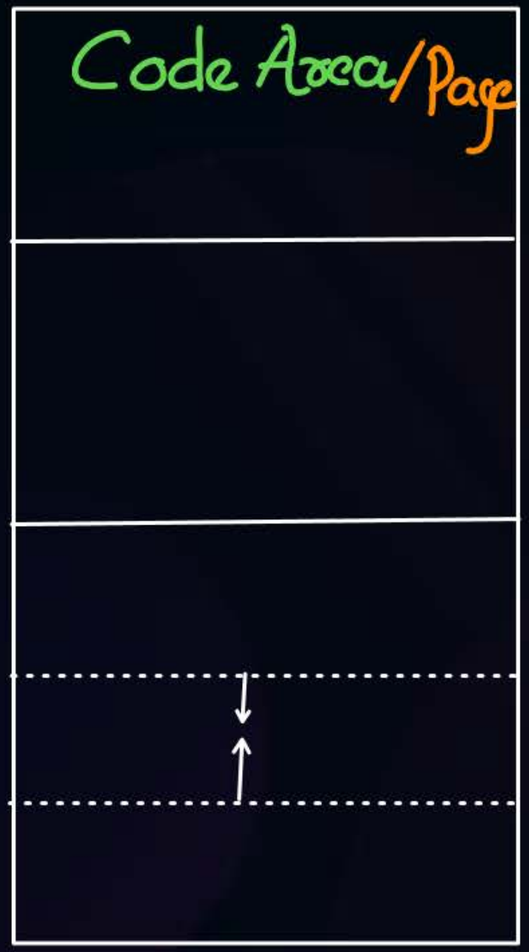
Local variables are stored in stack

Activation Record

Slide



Storage class



Slide



Storage class

Storage Class : The area of Memory where variable is stored

2 Scope of a variable.

3. Life time of a variable.

4 Inital value of a variable.

Slide



Storage class

In C-Language 4 storage class

1. auto ✓
2. static ✓
3. register
4. extern

Slide



Storage class

auto . auto short form of automatic

auto is default storage class of local variable

```
int main(){
    int a,
    .
}
```

```
int main(){
    auto int a;
```

Slide



Storage class

2 auto variable stored in stack

3 As soon as function start executing memory allocated for local variable & upon termination of function memory deallocated.

4 Initial value can be anything & called as garbage

5 Scope is within the function & Life time as long as function is in execution.

Slide



Storage class

Static : static is default Storage class for global variable

```
#include<stdio.h>
int a;
int main(){
.
.
.
}
```

```
#include<stdio.h>
static int a;
int main(){
.
.
.
}
```

Slide



Storage class

2 The static variable stored in Initialized data segment

3 Initial value for static variable is zero

4. Lifetime of static variable is as long as program is running

5. Local variable can also declared as static

Slide



Storage class

6. Local variable declared as static then it preserves its value during multiple function call.

7 Local static variable Initialized only once

Slide



Storage class

```
#include<stdio.h>
void fun() {
    static int a=10;
    a=a+10;
    printf("%d",a);
}
```

```
int main(){
    fun();
    fun();
    return 0;
}
```

a | 10 | 20

first print :- 20

Second call :- 30

preserve the last value

Slide



Storage class

Register :- Register is a memory within the processor

Because Register inside the processor hence we can Access them in faster manner.

If a variable of a program required faster access the we can request Compiler to allocate register.

Slide



Storage class

* for any Local variable only Local variable can have register storage class

```
int main()
 register int a;
 :
 :
}
```



Storage class

Extern : Extern is short form of external

extern keywords extends Scope of variable

using extern we can access global variable

defined in different file.



2 mins Summary

Topic — Memory Layout

Topic — auto

Topic — static

Topic — register

Topic — extern

Slide

THANK - YOU

